13 The PN Junction Diode

13.1 Basic Structure of the PN Junction

A PN junction is formed when a p-type and an n-type semiconductor are brought into contact to form a single-crystal structure. Figure 1 shows a simplified geometry of the PN junction, and Figure 2 illustrates the space charge region and the electric field that forms at the junction.

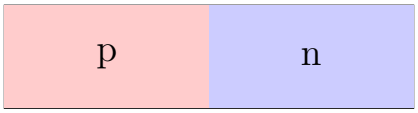


Figure 1: Simplified geometry of a Metallurgical PN junction

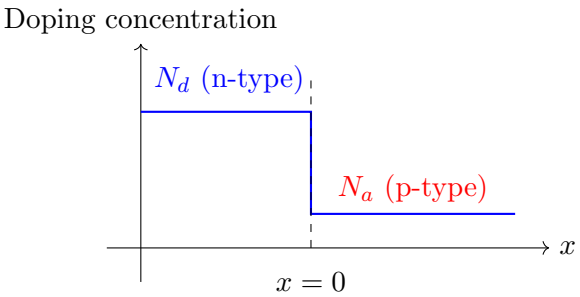


Figure 2: Doping profile of an ideal uniformly doped PN junction

13.2 Zero Applied Bias (Thermal Equilibrium)

When a PN junction is formed, majority carrier electrons in the n region diffuse into the p region, and majority carrier holes in the p region diffuse into the n region. This diffusion leaves behind ionized donors in the n region (positive charge) and ionized acceptors in the p region (negative charge), creating a **space charge region** (also called the **depletion region**) near the metallurgical junction. The electric field created by these fixed charges opposes further diffusion, establishing thermal equilibrium.

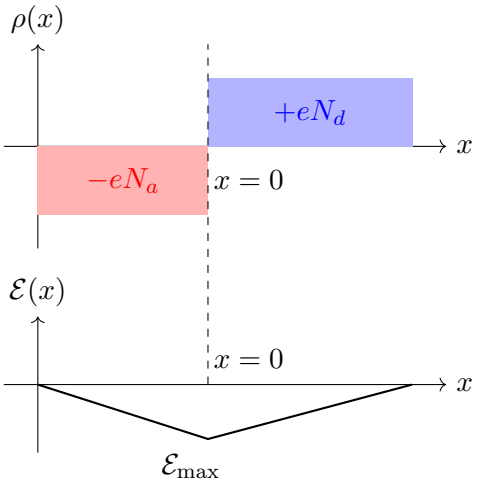


Figure 3: Space charge density and electric field in a uniformly doped PN junction

Energy Band Diagram and Built-in Potential

Figure 4 shows the energy band diagram of a PN junction in thermal equilibrium. The Fermi level is constant throughout the structure, and the conduction and valence bands bend through the space charge region. The built-in potential barrier V_{bi} is the potential difference across the junction that maintains equilibrium.

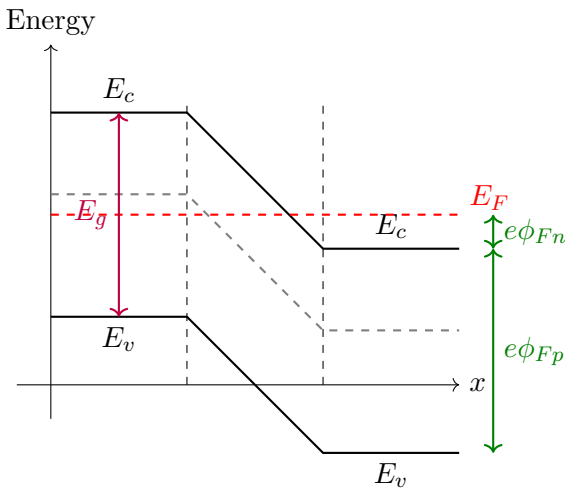


Figure 4: Energy band diagram of a PN junction in thermal equilibrium

Derivation of the Built-in Potential

The built-in potential barrier V_{bi} can be determined from the difference between the intrinsic Fermi levels in the p and n regions. From earlier chapters, we have:

$$n_0 = n_i \exp\left(\frac{E_F - E_{Fi}}{kT}\right), \qquad p_0 = n_i \exp\left(\frac{E_{Fi} - E_F}{kT}\right)$$

For the n region, with $n_0 = N_d$, we define:

$$e\phi_{Fn} = E_{Fi} - E_F \quad \Rightarrow \quad N_d = n_i \exp\left(-\frac{e\phi_{Fn}}{kT}\right)$$

Solving for ϕ_{Fn}:

$$\phi_{Fn} = -\frac{kT}{e} \ln\left(\frac{N_d}{n_i}\right)$$

For the p region, with $p_0 = N_a$, we define:

$$e\phi_{Fp} = E_{Fi} - E_F \quad \Rightarrow \quad N_a = n_i \exp\left(\frac{e\phi_{Fp}}{kT}\right)$$

Solving for ϕ_{Fp}:

$$\phi_{Fp} = \frac{kT}{e} \ln\left(\frac{N_a}{n_i}\right)$$

The built-in potential is the sum of the magnitudes of these two potentials:

$$\boxed{V_{bi} = |\phi_{Fn}| + |\phi_{Fp}| = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)}$$

where $V_t = kT/e$ is the thermal voltage (approximately 0.0259 V at 300 K).

Example 6 (Built-in Potential Calculation)**.** Consider a silicon PN junction at $T = 300$ K with doping concentrations $N_a = 2 \times 10^{17}$ cm^{-3} and $N_d = 10^{15}$ cm^{-3}. The intrinsic carrier concentration for silicon is $n_i = 1.5 \times 10^{10}$ cm^{-3}. Calculate the built-in potential.

Solution:

$$V_{bi} = V_t \ln\left(\frac{N_a N_d}{n_i^2}\right) = (0.0259) \ln\left[\frac{(2 \times 10^{17})(10^{15})}{(1.5 \times 10^{10})^2}\right] = 0.713 \text{ V}$$

If the doping in the p region is reduced to $N_a = 10^{16}$ cm^{-3}, then:

$$V_{bi} = 0.0259 \ln \left[\frac{(10^{16})(10^{15})}{(1.5 \times 10^{10})^2} \right] = 0.635 \text{ V}$$

Comment: The built-in potential changes only slightly as doping concentrations change by orders of magnitude due to the logarithmic dependence.

13.3 Electrostatics of the Space Charge Region

The electric field and potential in the space charge region are determined by solving Poisson's equation. For a uniformly doped abrupt junction, the charge density is:

$$\rho(x) = -eN_a \quad (-x_p < x < 0), \qquad \rho(x) = +eN_d \quad (0 < x < x_n)$$

Integrating Poisson's equation yields the electric field distribution (linear through the junction) and the potential distribution (quadratic). The key results are:

- **Charge neutrality condition:** $N_a x_p = N_d x_n$
- **Maximum electric field (at $x = 0$):** $|\mathcal{E}_{\max}| = \dfrac{eN_a x_p}{\epsilon_s} = \dfrac{eN_d x_n}{\epsilon_s}$
- **Built-in potential:** $V_{bi} = \dfrac{e}{2\epsilon_s}(N_a x_p^2 + N_d x_n^2)$

The space charge widths extending into each region are:

$$\boxed{x_n = \left[\frac{2\epsilon_s V_{bi}}{e} \left(\frac{N_a}{N_d} \right) \left(\frac{1}{N_a + N_d} \right) \right]^{1/2}}, \qquad \boxed{x_p = \left[\frac{2\epsilon_s V_{bi}}{e} \left(\frac{N_d}{N_a} \right) \left(\frac{1}{N_a + N_d} \right) \right]^{1/2}}$$

The total space charge width is:

$$\boxed{W = x_n + x_p = \left[\frac{2\epsilon_s V_{bi}}{e} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2}}$$

where:

- ϵ_s = semiconductor permittivity ($\epsilon_s = \epsilon_r \epsilon_0$)
- e = elementary charge (1.602×10^{-19} C)
- N_a, N_d = acceptor and donor concentrations (cm^{-3})
- V_{bi} = built-in potential (V)

For a detailed derivation, students are referred to Neamen's *Semiconductor Physics and Devices*, Chapter 7.

13.4 Reverse Bias

When a positive voltage V_R is applied to the n region with respect to the p region (reverse bias), the potential barrier increases. The total barrier becomes $V_{bi} + V_R$. The space charge width increases as:

$$\boxed{W = \left[\frac{2\epsilon_s (V_{bi} + V_R)}{e} \left(\frac{N_a + N_d}{N_a N_d} \right) \right]^{1/2}}$$

The space charge widths in each region increase by the same factor:

$$x_n(V_R) = x_n(0)\sqrt{1 + \frac{V_R}{V_{bi}}}, \qquad x_p(V_R) = x_p(0)\sqrt{1 + \frac{V_R}{V_{bi}}}$$

The maximum electric field also increases:

$$|\mathcal{E}_{\max}| = \frac{2(V_{bi} + V_R)}{W}$$

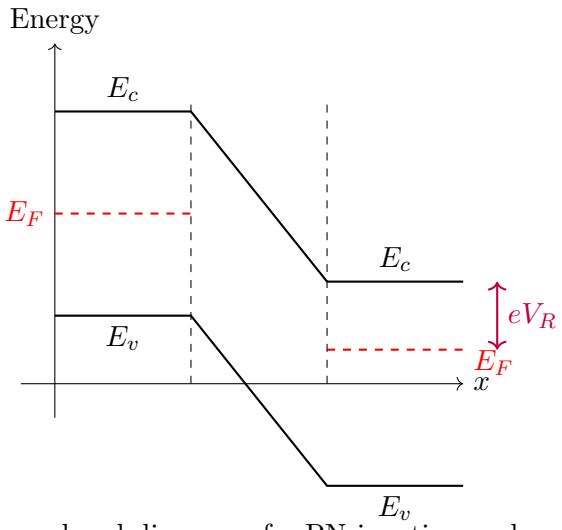


Figure 5: Energy band diagram of a PN junction under reverse bias

13.5 Junction Capacitance

The separation of fixed charges in the depletion region creates a capacitance. When the reverse bias voltage changes by dV_R, the charge stored in the depletion region changes by dQ', giving rise to the **junction capacitance** (also called the **depletion layer capacitance**):

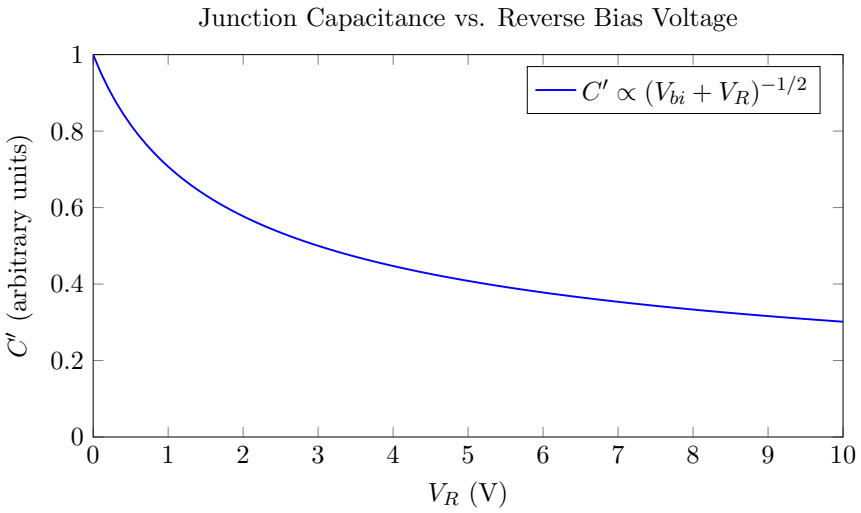
$$\boxed{C' = \frac{dQ'}{dV_R} = \left[\frac{e\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)}\right]^{1/2}}$$

A useful alternative expression shows that the junction capacitance behaves like a parallel plate capacitor:

$$\boxed{C' = \frac{\epsilon_s}{W}}$$

For a one-sided junction (e.g., p^+n where $N_a \gg N_d$), the capacitance simplifies to:

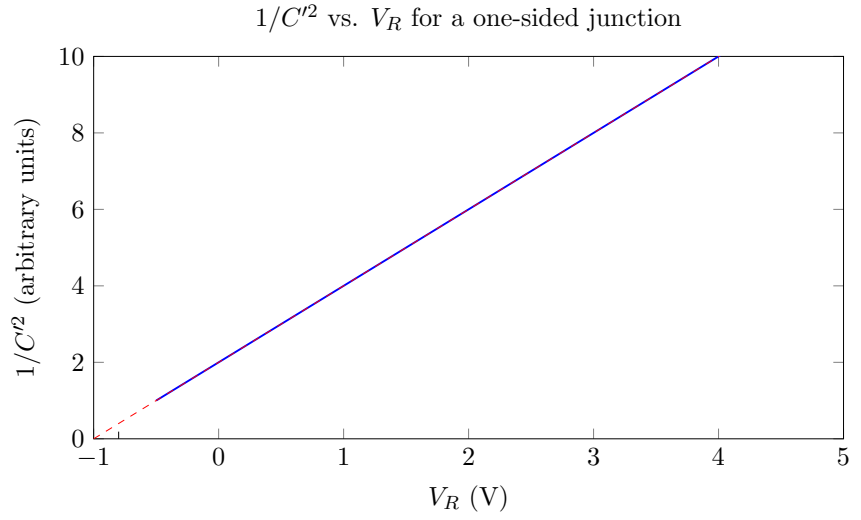
$$\boxed{C' \approx \left[\frac{e\epsilon_s N_d}{2(V_{bi} + V_R)}\right]^{1/2}}$$



Importance of Junction Capacitance

The junction capacitance has several important practical implications:

- **High-frequency operation:** In PN junction diodes (e.g., switching diodes, rectifiers), the junction capacitance limits the maximum operating frequency. When the diode is reverse-biased, the depletion region acts as a capacitor that must be charged and discharged during switching.
- **Voltage-controlled capacitors:** The strong dependence of C' on V_R is exploited in **varactor diodes** (varicap diodes). By varying the reverse bias voltage, the capacitance can be tuned. This property is used in:
  - Voltage-controlled oscillators (VCOs) for frequency modulation
  - Electronic tuning circuits (e.g., radio tuners)
  - Phase-locked loops (PLLs) and frequency synthesizers
- **Device modeling:** The junction capacitance is an essential parameter in SPICE models for diodes and transistors, affecting transient response and AC performance.
- **Characterization of doping profiles:** Measuring C' as a function of V_R allows experimental determination of doping concentrations. A plot of $1/C'^2$ vs. V_R yields a straight line whose slope gives N_d (for a one-sided junction) and whose intercept gives V_{bi}.



Summary of Key Parameters

Parameter	**Expression**
Built-in potential (V_{bi})	$V_{bi} = \frac{kT}{e} \ln\left(\frac{N_a N_d}{n_i^2}\right)$
Space charge width (zero bias)	$W = \left[\frac{2\epsilon_s V_{bi}}{e}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2}$
Space charge width (reverse bias)	$W = \left[\frac{2\epsilon_s (V_{bi} + V_R)}{e}\left(\frac{N_a + N_d}{N_a N_d}\right)\right]^{1/2}$
Junction capacitance	$C' = \frac{\epsilon_s}{W} = \left[\frac{e\epsilon_s N_a N_d}{2(V_{bi} + V_R)(N_a + N_d)}\right]^{1/2}$
One-sided junction (p^+n)	$C' \approx \left[\frac{e\epsilon_s N_d}{2(V_{bi} + V_R)}\right]^{1/2}$

The material presented here establishes the electrostatic foundation of the PN junction. In the next chapter, we will extend this analysis to consider current flow under forward bias and derive the diode current-voltage characteristics.